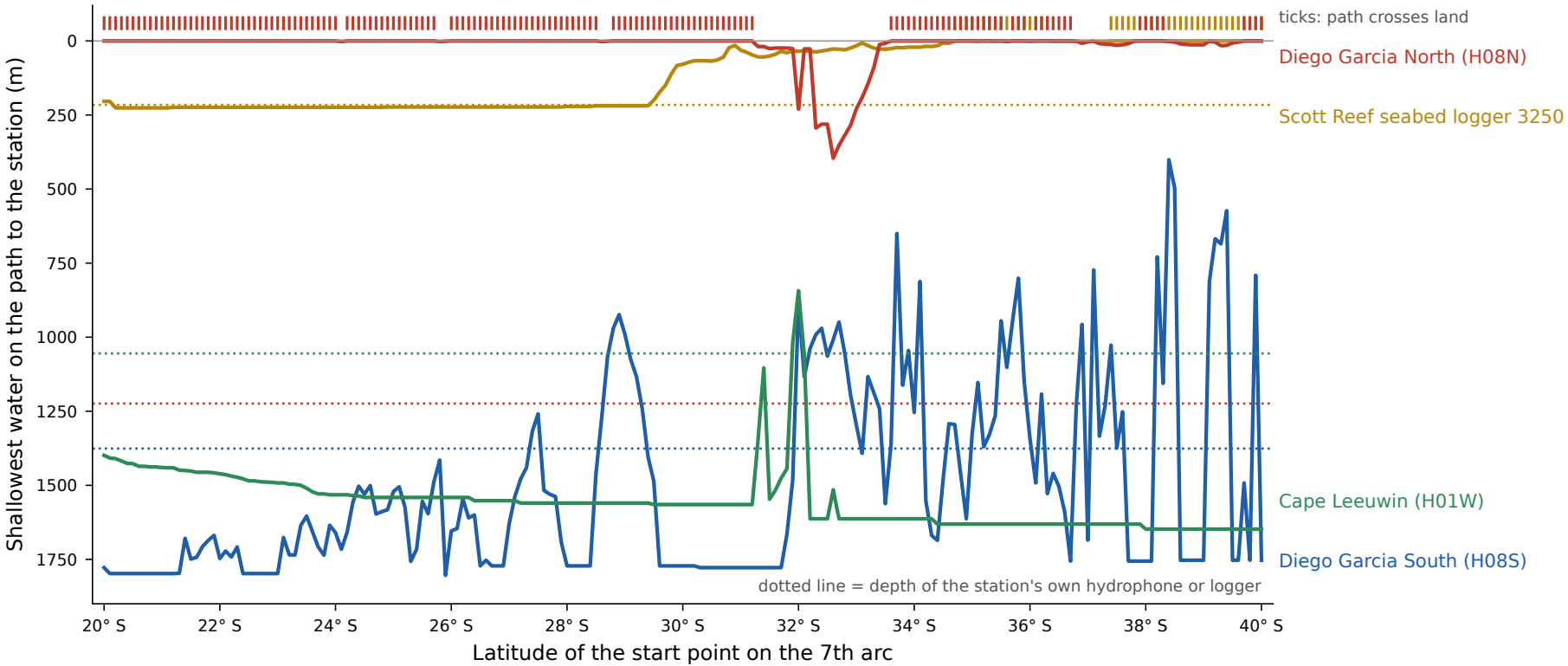

Plain note: Each line shows the shallowest water on the straight path from a point on the 7th arc to a listening station. A tick at the top shows a path that crosses land or reef above sea level. Sound in the deep sound channel cannot pass land and loses much energy over water much shallower than about 1,000 m. Diego Garcia North: every path crosses the Chagos reefs or water of 400 m or less, so this station is dropped. Scott Reef: from 20° S to about 29.5° S no point on the path is shallower than the logger itself (216 m), so this station is not dropped there. South of 29.5° S the path crosses the North West Shelf, and from about 35° S it crosses land.

Technical note: 7th arc: small circle, centre 0.570° N 64.252° E, radius 44.501° (fit to 00:19:29 positions), checked against the sea-level ring (36° S at 91.4° E; 28° S at 99.8° E; agreement within 0.13° of longitude); 201 start points at 0.1° steps. Paths: WGS84 geodesics (pyproj), sampled every 250 m; GEBCO_2026 15 arc-second elevation (local copy), nearest cell. The shallowest point excludes the last 2 km before the receiver. Station positions: IMS triad centroids from FDSN (EarthScope) station metadata; Scott Reef 3250 from the IMOS deployment metadata (15° 29.002' S, 121° 15.060' E, 216 m). Hydrophone depths: H08N 1,182–1,248 m, H08S 1,356–1,413 m, H01W 1,046–1,063 m. Scott Reef, 20–29.4° S: shallowest point 200–226 m, located 2–14 km from the logger (its own slope); before the last 100 km the shallowest point is 500–616 m and 17–590 km of path lies shallower than 1,000 m (Exmouth Plateau region), so high propagation loss is expected but there is no blockage. H08N: shallowest point −198 m (land) to 396 m at every start point; reefs and banks of the Chagos Archipelago lie 30–200 km from the hydrophones. Bathymetry only; no propagation model in this chart.